

Ryan Vincent

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London, UK

EDUCATION

Imperial College London <i>M.Eng. Computer Science</i> • Grade: Upper Second-Class Honours (Current)	London, UK 2023 – 2027
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EXPERIENCE

Howden Insurance Brokers <i>Intern Data Engineer</i> • Engineered robust ETL modules using Azure Data Factory to ingest unstructured data from Data Lakes and optimize schemas for the Enterprise Data Warehouse (EDW). • Orchestrated CI/CD pipelines within an Agile team of 8, automating testing and deployment processes.	London, UK Nov 2023 – Jun 2024
Haut Technologies Inc. <i>Intern</i> • Built a computer vision model using Keras and TensorFlow to quantify acne severity and classify dermatological features with high accuracy.	London, UK Jul 2022 – Aug 2022
University of Southampton <i>Apprentice</i> • Developed a respiration analyser on the Arduino microcontroller using machine learning in Python, performing real-time signal acquisition and data analysis.	Southampton, UK Jun 2021 – Sep 2021

PROJECTS & ACHIEVEMENTS

Algorithmic Trading System (IMC Prosperity) <i>Python, Bayesian Optimization, Pandas</i> • Award: Ranked 3rd in the UK in the IMC Prosperity Global Trading Challenge. • Engineered a high-frequency trading algorithm in Python to trade virtual commodities on a simulated electronic exchange, utilizing z-score mean reversion strategies. • Developed a Bayesian optimization framework to dynamically tune strategy hyperparameters (e.g., moving average windows, entry/exit thresholds), which maximized profit significantly compared to static models. • Implemented statistical arbitrage and market-making logic to capitalize on spread inefficiencies.	April 2024
Map.it (Citadel ICHack 2024) <i>React, Python, Google Maps API</i> • Award: Won Newcomer Category Award; Chosen as Runner-up by Citadel. • Architected a location intelligence tool for start-ups to identify optimal business locations based on economic indicators. • Integrated Google Maps API to aggregate commercial real estate data and transit costs, processing inputs through a custom profitability algorithm and GPT-4 for summaries.	Jan 2024
Carbon Footprint Tracker (CISCO ICHack 2023) <i>JavaScript, Browser Extension</i> • Award: Runner-up in product development. • Engineered a browser extension that estimates real-time carbon emissions by analyzing network packet size and data center energy profiles.	Jan 2023
Noise Pollution Classifier <i>Python, Raspberry Pi, CNN</i> • Award: Won Global First Place in Apps for Good Competition. • Developed a portable, real-time Raspberry Pi solution to analyze environmental noise and identify pollution sources. • Captured acoustic data to generate spectrograms, training a CNN to classify pollution sources via Softmax.	Jan 2020 – Jun 2020

TECHNICAL SKILLS

Languages: Python, Java, Haskell, Kotlin, Rust, JavaScript, SQL, PostgreSQL, C++, ARM Assembly
Frameworks & Libraries: Keras, PyTorch, TensorFlow, React, Pandas, NumPy
Cloud & DevOps: AWS, Azure DevOps, Git, Docker, Azure Data Factory
Concepts: Machine Learning, Data Structures & Algorithms, ETL, Agile, Quantitative Analysis